

FHIRLab Core — APAC Cloud Hosting Cost Estimate

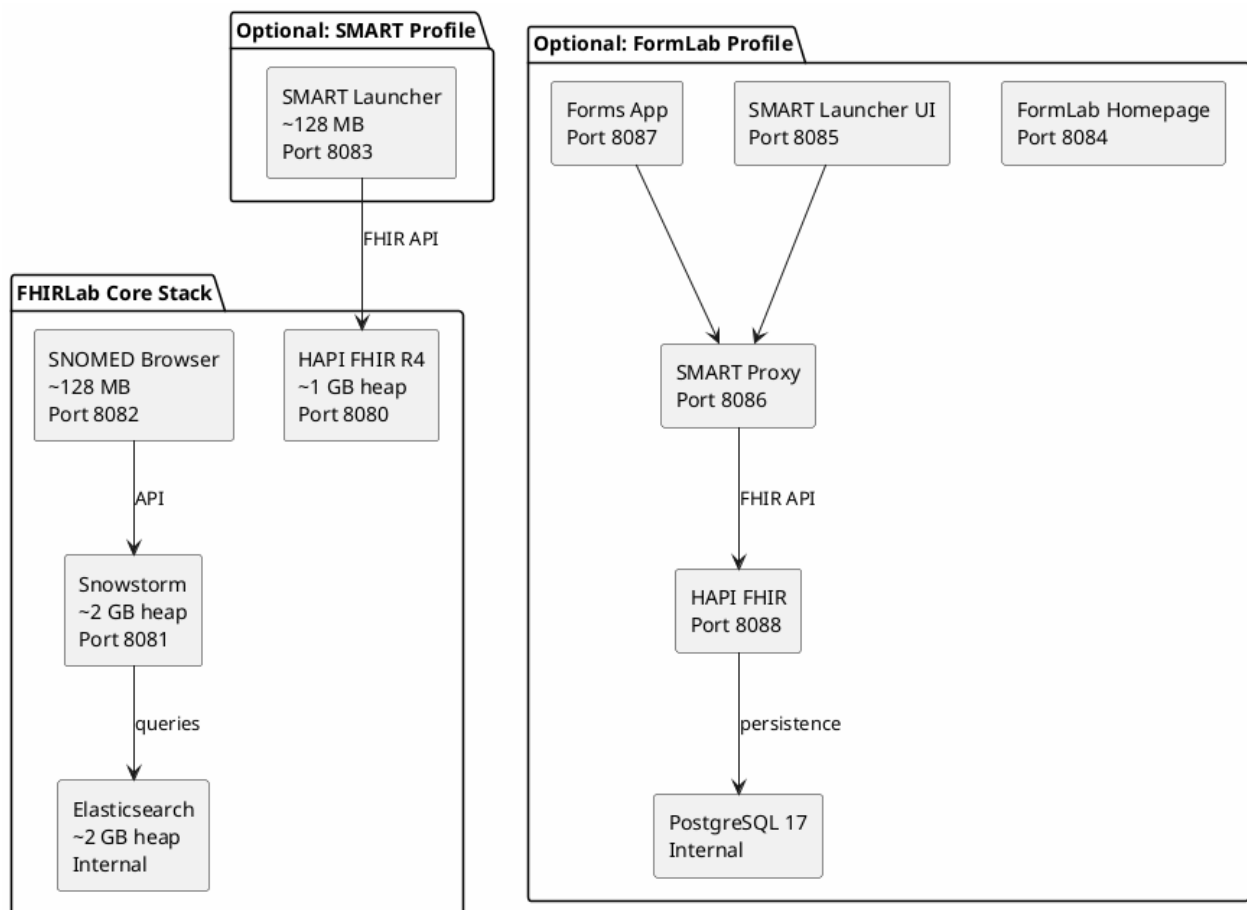
Date: 2026-04-27 **Status:** Draft **GitLab Issue:** [gitlab-profile#26](#)

1. Scope

Estimate the monthly operating cost of deploying FHIRLab Core on a low-cost cloud provider in the APAC region, so regional teams can understand the financial commitment before adoption.

Stack Under Evaluation

The core Docker Compose stack consists of:



Resource Requirements

Resource	Core Stack	+ SMART	+ FormLab	All Profiles
RAM	~5.5 GB	~5.6 GB	~7.5 GB	~7.6 GB
vCPU	2-4	2-4	4	4
Storage	50 GB	50 GB	60 GB	60 GB

Resource	Core Stack	+ SMART	+ FormLab	All Profiles
Bandwidth	< 1 TB/mo	< 1 TB/mo	< 1 TB/mo	< 1 TB/mo

Minimum viable instance: 4 vCPU, 8 GB RAM, 80-160 GB SSD. The README states ≥ 4 GB RAM, but Elasticsearch alone requires 2 GB heap plus Snowstorm at 2 GB, making 8 GB the practical minimum.

Deployment Constraint

Per the SSCP DevOps Catchup (2026-04-17): deployment is scoped to **localhost / LAN access** using Docker Compose only. Public DNS, IP addresses, and TLS certificates are **out of scope**.

2. Provider Comparison

Monthly Cost (4 vCPU / 8 GB+ RAM)

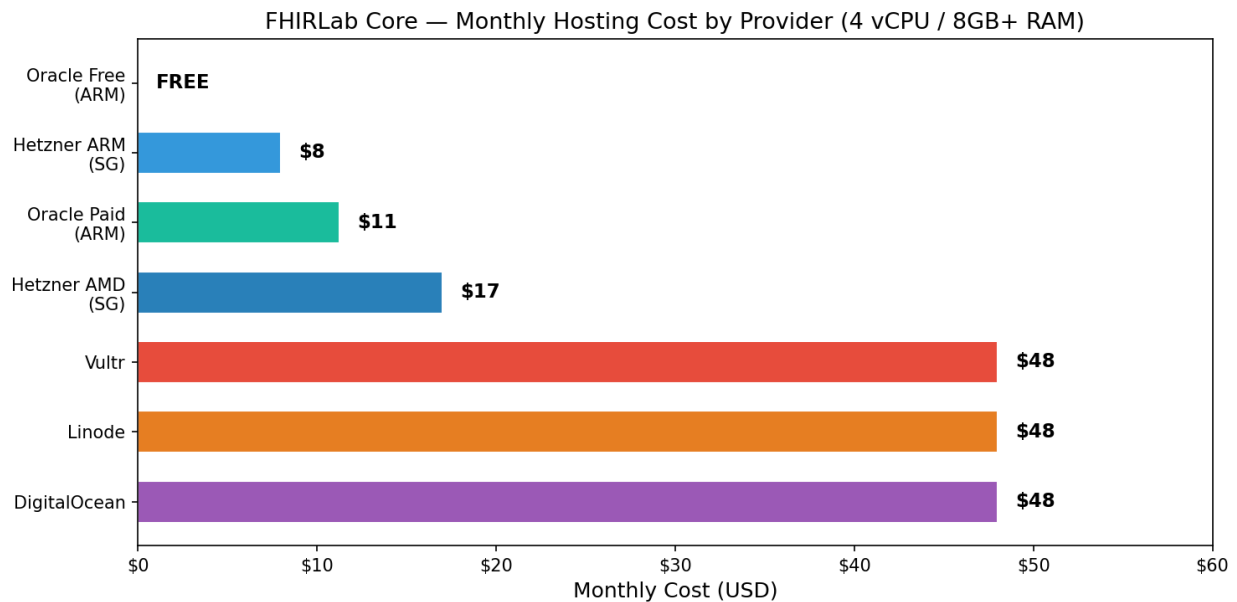


Figure 1: Monthly Cost Comparison

Provider	Plan	Monthly	Storage	Bandwidth	Excess Egress	APAC Regions
Oracle Free	A1.Flex 4 OCPU / 24 GB ARM	\$0	200 GB	10 TB	\$0.0085/GB	SG, TYO, SYD, MUM + 4 more
Hetzner ARM	CAX21 4 vCPU / 8 GB ARM	~\$8	80 GB	20 TB	~\$0.001/GB	Singapore only
Oracle Paid	A1.Flex 4 OCPU / 8 GB ARM	~\$10	+\$1.25	10 TB	\$0.0085/GB	SG, TYO, SYD, MUM + 4 more
Hetzner AMD	CPX31 4 vCPU / 8 GB	~\$17	160 GB	20 TB	~\$0.001/GB	Singapore only
Vultr	Cloud Compute 4 vCPU / 8 GB	\$48	160 GB	5 TB	\$0.01/GB	SG, TYO, SYD, MUM
Linode	Shared 4 vCPU / 8 GB	\$48	160 GB	5 TB	\$0.01/GB	SG, TYO, SYD, MUM
DigitalOcean	Regular 4 vCPU / 8 GB	\$48	160 GB	5 TB	\$0.01/GB	SG, SYD, BLR

Resource Comparison

3-Year Cumulative Cost

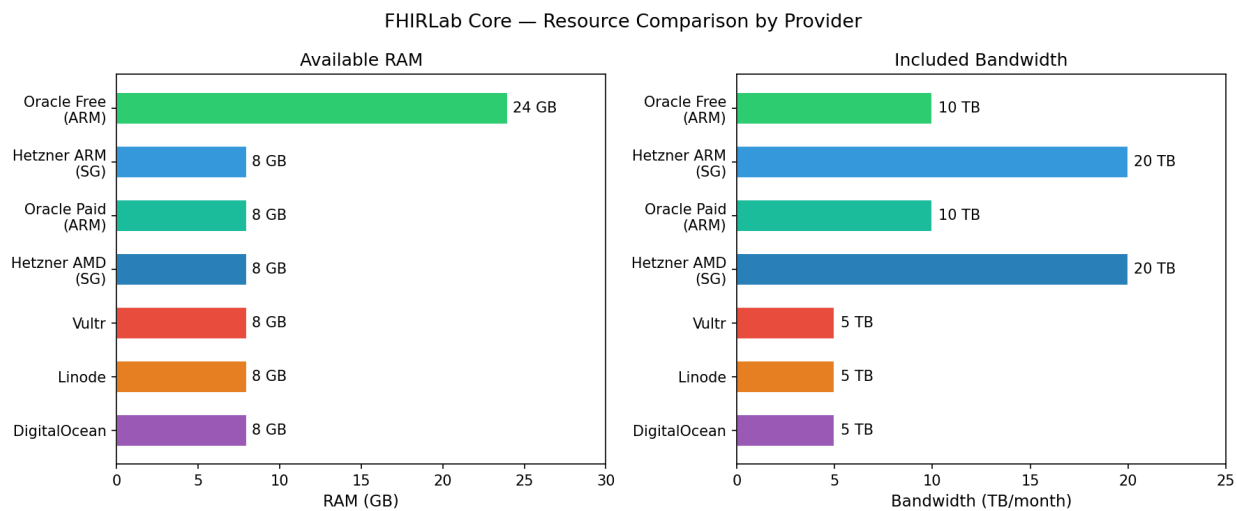


Figure 2: Resource Comparison

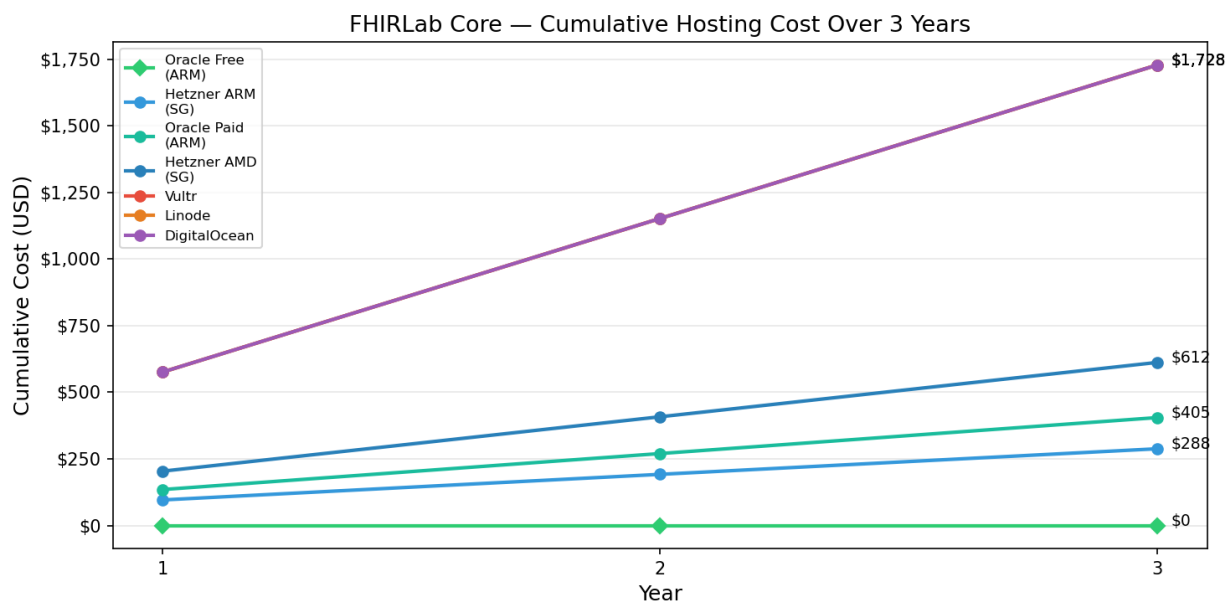
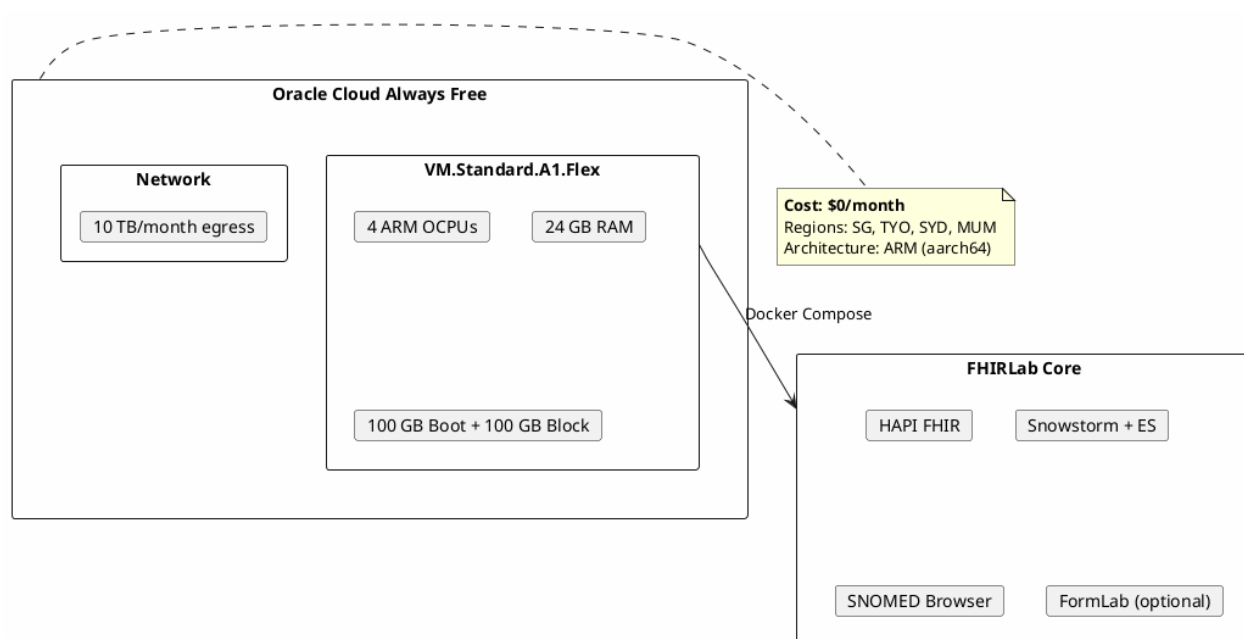


Figure 3: Annual Cost Projection

Provider	Year 1	Year 2	Year 3
Oracle Free	\$0	\$0	\$0
Hetzner ARM	\$96	\$192	\$288
Oracle Paid	\$135	\$270	\$405
Hetzner AMD	\$204	\$408	\$612
Vultr / Linode / DO	\$576	\$1,152	\$1,728

3. Recommendation

Primary: Oracle Cloud Free Tier (ARM)



Why Oracle Free Tier?

- **\$0/month** — genuinely free, not a trial
- **24 GB RAM** — 3× the minimum, ample headroom for all profiles including Form-Lab
- **200 GB storage** — more than sufficient
- **8 APAC regions** — best coverage for target countries (Philippines, Fiji, Malaysia)
- **10 TB bandwidth** — 10× expected usage

Caveats:

1. **ARM architecture** — all Docker images in the stack (HAPI FHIR, Elasticsearch, Snowstorm, PostgreSQL) publish multi-arch images including linux/arm64. Verified:
 - hapiproject/hapi — multi-arch ✓
 - docker.elastic.co/elasticsearch/elasticsearch:8.11.1 — multi-arch ✓

- `snomedinternational/snowstorm` — multi-arch ✓
 - `postgres:17` — multi-arch ✓
2. **Capacity constraints** — Oracle Free Tier ARM instances can be difficult to provision in popular regions due to demand. Retry or try a less popular region (e.g. Osaka, Hyderabad, Melbourne).
 3. **No SLA** — free tier has no uptime guarantee. Acceptable for a learning sandbox.

Fallback: Hetzner Cloud Singapore

If Oracle Free Tier is unavailable:

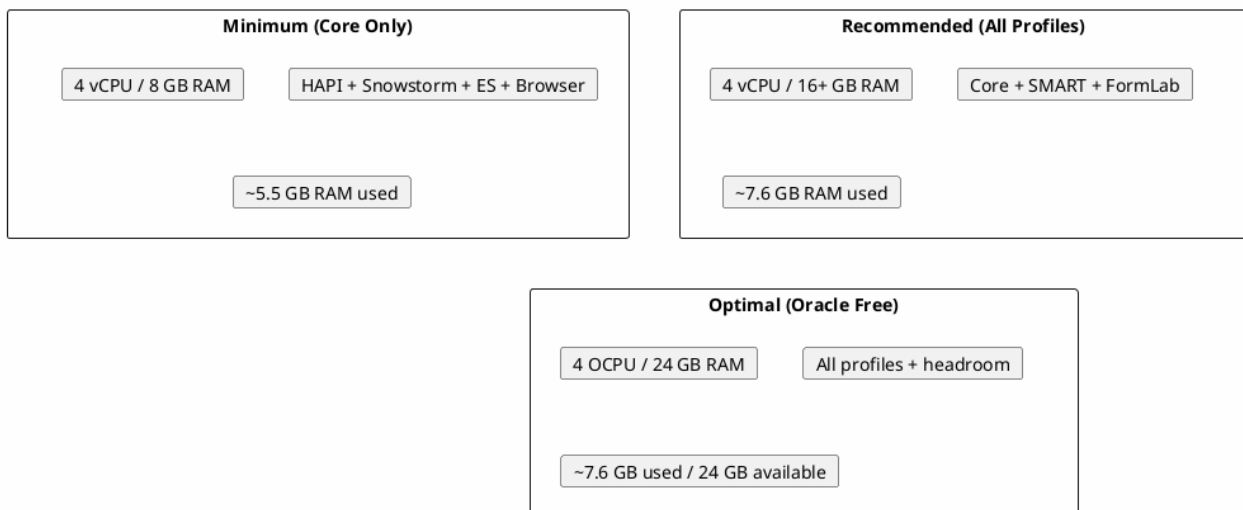
Option	Monthly	Notes
CAX21 (ARM)	~\$8	4 vCPU / 8 GB / 80 GB / 20 TB BW
CPX31 (AMD)	~\$17	4 vCPU / 8 GB / 160 GB / 20 TB BW

Hetzner is 65–85% cheaper than Vultr/Linode/DigitalOcean. Limited to Singapore in APAC, but Singapore provides reasonable latency to the Philippines, Malaysia, and broader Southeast Asia.

Not Recommended for This Use Case

Vultr, Linode, and DigitalOcean at \$48/month are **3–6× more expensive** than alternatives for equivalent specs. They offer broader APAC presence and mature ecosystems, but the cost premium is not justified for a learning sandbox with no SLA requirement.

4. Deployment Sizing by Profile



5. Summary

	Oracle Free	Hetzner ARM	Hetzner AMD	Vultr/Linode/DO
Monthly cost	\$0	~\$8	~\$17	\$48
Annual cost	\$0	~\$96	~\$204	\$576
3-year cost	\$0	~\$288	~\$612	\$1,728
RAM	24 GB	8 GB	8 GB	8 GB
APAC regions	8	1 (SG)	1 (SG)	3-4
Architecture	ARM	ARM	x86	x86
SLA	None	Standard	Standard	Standard
Suitable for sandbox	Yes	Yes	Yes	Yes (overpriced)

Bottom line: Oracle Cloud Free Tier at **\$0/month** is the recommended choice. Hetzner Singapore at **\$8-17/month** is the best paid alternative. The \$48/month providers offer no meaningful advantage for this use case.

References

- [Oracle Cloud Free Tier](#)
- [Hetzner Cloud Pricing](#)
- [Vultr Pricing](#)
- [Linode/Akamai Pricing](#)
- [DigitalOcean Pricing](#)
- FHIRLab Core [README](#) and [docker-compose.yml](#)

Note: Prices reflect publicly listed rates as of April 2026. Verify current pricing at each provider's website before committing.